7-Qubit Planar Superconducting Transmon Processor: Guided Design Manual and Laboratory Engineering Chronicle

Atharve Dahima

Abstract—This living engineering chronicle provides a fully reproducible, step-by-step guided manual for designing a planar 7-qubit superconducting transmon quantum processor on silicon. We avoid detached theoretical proofs; instead, every decision is justified by physical and microwave constraints encountered on the lab bench. The chronicle details: (1) Operating temperature limits (15 mK) and cryogenic attenuation cascades; (2) The breakdown of linear LC tanks and the necessity of Josephson non-linearity; (3) The $50.0\,\Omega$ impedance standard and coplanar waveguide geometry; (4) Analytical derivation of target lumped parameters ($f_q \approx 5.0$ GHz, $f_r \approx 7.04$ GHz, $C_q = 70$ fF, $L_J = 10$ nH, $C_r = 340$ fF, $L_r = 1.5$ nH); and (5) Circuit synthesis in Ansys Circuit (Nexxim). This document is updated incrementally as hardware simulations progress, serving as a standalone reference from initial schematics to mask tape-out.

Index Terms—Circuit QED, Transmon Qubit, Josephson Junction, Anharmonicity, Coplanar Waveguide, 50-Ohm Matching, Lumped Synthesis, Ansys Circuit, S-Parameters.

1 System Architecture and Environmental Bounds

1.1 Why Superconducting Circuits?

Building a quantum processor requires two competing properties: isolation from noise to protect delicate superpositions, and strong coupling to control lines to execute fast logic gates. Trapped ions and neutral atoms offer long coherence, but optical switching limits their two-qubit gate speeds to microseconds. Solid-state planar circuits solve the speed problem.

Lithographically patterned aluminum and niobium on high-resistivity silicon behave as artificial atoms. Operating at microwave frequencies (4–8 GHz), these devices respond to nanosecond control pulses (10–30 ns). They interface directly with established microwave Electronic Design Automation (EDA) suites: Ansys Electronics Desktop (AEDT), Qiskit Metal, and KLayout.

A. Dahima leads the design and simulation of this superconducting quantum processor. Lab repository: https://github.com/atharveeee-netizen/7-qubit-superconducting-processor. Inquiries: atharveeee@gmail.com.

Comparison of Quantum Computing Modalities

Superconducting Circuits (Our Architecture):
- Monolithic planar thin-film lithography (Nb/Al on Si)
- Nanosecond-scale gate execution (10–30 ns clock cycle)
- Direct integration with high-frequency microwave EDA tools

Atomic / Trapped-Ion Systems:
- Vacuum chamber / laser overhead; slower gate times (μs range)

Fig. 1. Architectural advantages of planar superconducting circuit QED.

1.2 Why Must the Processor Sit at 15 mK?

At room temperature (300 K), ambient thermal energy is roughly 25.8 meV. A single microwave photon at our operating frequency (5.0 GHz) carries far less energy:

$$E_{\text{photon}} = hf_{01} \approx 20.7\,\mu\text{eV} \tag{1}$$

If left at room temperature, blackbody thermal radiation swamps the qubit, causing rapid spontaneous excitations. Freezing the device into its ground state $|0\rangle$ requires $k_BT \ll hf_{01}$.

Inside a ^{3}He/^{4}He dilution refrigerator operating at 15 mK, thermal energy drops to $k_BT \approx 1.29\,\mu$eV. The Boltzmann probability of finding the qubit in its excited state without an intentional control drive falls to:

$$P_1 = \exp\left(-\frac{hf_{01}}{k_BT}\right) \approx \exp\left(-\frac{20.7}{1.29}\right) \approx 1.1 \times 10^{-7} \tag{2}$$

Thermal excitations become virtually nonexistent, ensuring deterministic state initialization.

1.3 Why Install Drive Line Attenuators?

Even when the dilution refrigerator base plate reaches 15 mK, drive cables linked to room-temperature signal generators route 300 K Johnson-Nyquist thermal noise down the line. Unchecked, this noise reaches the sample holder and destroys coherence.

To prevent this, fixed attenuators are anchored at each cooling stage:

- **4 K Stage (Still):** -20 dB attenuator strips high-temperature thermal noise.
- **100 mK Stage (Cold Plate):** -10 dB attenuator provides intermediate thermalization.

- **15 mK Stage (Mixing Chamber):** -30 dB attenuator terminates residual photons into cold metal, ensuring only coherent microwave control tones reach the chip.

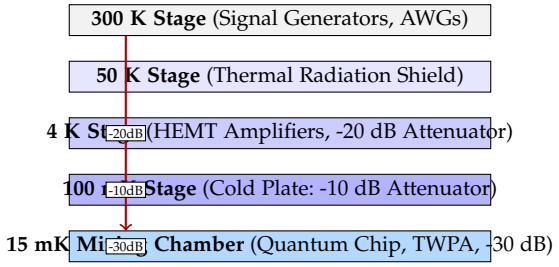


Fig. 2. Thermal staging inside the dilution refrigerator showing drive line attenuation.

2 Physical Circuit Foundations

2.1 Why an LC Tank Cannot Serve as a Qubit

A standard LC resonator has a parabolic potential $U(\Phi) = \Phi^2/(2L)$. This geometry creates equidistant energy rungs:

$$\Delta E_{0\to1} = \Delta E_{1\to2} = \Delta E_{2\to3} = hf_0 \tag{3}$$

When a microwave pulse drives the $|0\rangle \to |1\rangle$ transition, the identical energy gap causes the state to leak into $|2\rangle, |3\rangle$, and higher levels. An LC tank is a harmonic oscillator, not a controllable two-level qubit.

2.2 Why Use a Josephson Junction?

Replacing the linear inductor with a sub-micron Josephson junction changes the potential from parabolic to cosinusoidal:

$$U(\phi) = -E_J \cos\phi \approx \frac{1}{2}E_J\phi^2 - \frac{1}{24}E_J\phi^4 \tag{4}$$

The negative quartic term $-\frac{1}{24}E_J\phi^4$ softens the potential well at higher energies, compressing upper rungs. This produces negative anharmonicity:

$$\alpha = f_{12} - f_{01} \approx -E_C/h \tag{5}$$

Because $f_{12} \neq f_{01}$, a control tone tuned to f_{01} cannot drive higher-level transitions.

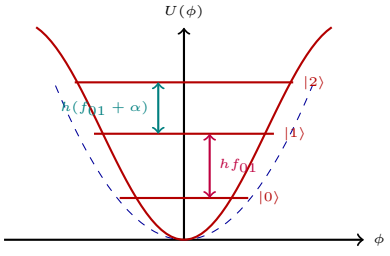


Fig. 3. Parabolic potential (linear inductor) vs. cosinusoidal potential (Josephson junction).

2.3 Why Target the Transmon Ratio $E_J/E_C \approx 50$?

In early Cooper-pair boxes ($E_J \approx E_C$), stray background charges (n_g) constantly shifted the qubit transition frequency, destroying phase coherence ($T_2^* < 1\,\mu$s).

By shunting the junction with large capacitor pads to enforce $E_J/E_C \gg 1$, charge sensitivity drops exponentially:

$$\epsilon_m \propto \exp\left(-\sqrt{8E_J/E_C}\right) \tag{6}$$

At $E_J/E_C \approx 50$, charge noise dephasing is suppressed by orders of magnitude, while preserving sufficient anharmonicity ($|\alpha| \approx 280$–330 MHz) to allow short 20 ns control pulses without state leakage.

2.4 Fabrication: Dolan Bridge Shadow Evaporation

Standard optical etching cannot define clean sub-micron junctions. Instead, we use double-angle Dolan bridge lithography:

1) Spin a bilayer resist (MMA bottom layer, PMMA top layer) onto high-resistivity silicon.
2) Electron-beam lithography exposes the pattern. MMA dissolves faster during development, creating an undercut beneath a suspended PMMA bridge.
3) Evaporate aluminum at angle $+\theta$ in an ultra-high vacuum chamber.
4) Bleed pure oxygen gas into the chamber ($P \approx 1$ Torr, $t \approx 10$ min) to grow an amorphous ~ 1 nm Al_2O_3 tunnel barrier.
5) Evaporate a second aluminum layer at angle $-\theta$, completing the overlap tunnel junction.

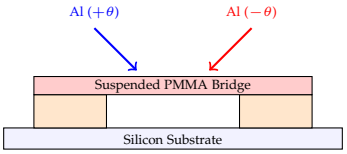


Fig. 4. Dolan bridge double-angle shadow evaporation cross-section.

3 Microwave Engineering Rationales

3.1 Why Match Impedance to Exactly $50.0\,\Omega$?

RF generators, digitizers, cryogenic coaxial cables, and HEMT amplifiers are manufactured to a $Z_0 = 50.0\,\Omega$ characteristic impedance. Any on-chip impedance jump produces a voltage reflection:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{7}$$

When reflections occur:

1) Microwave pulses bounce between bond wires and microwave packaging, creating standing waves that distort control pulse envelopes.
2) Impedance mismatches inject photon shot noise into the feedline, dephasing the qubit and lowering T_2^*.

On silicon ($\epsilon_r = 11.45$, $h = 500\,\mu$m), we achieve $Z_0 = 50.0\,\Omega$ using a Coplanar Waveguide (CPW) with center conductor width $W = 10\,\mu$m and gap spacing $S = 6\,\mu$m.

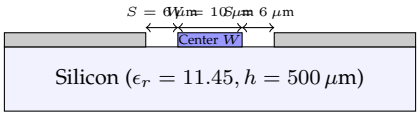


Fig. 5. Coplanar Waveguide (CPW) geometry providing $Z_0 = 50.0\,\Omega$.

3.2 Why Choose 5.0 GHz for the Qubit and 7.04 GHz for the Resonator?

Two constraints bound the frequency selection:

1) **Low-Frequency Limit ($f > 1$ GHz):** Operating frequencies must sit safely above thermal noise at 15 mK.
2) **High-Frequency Limit ($f < 80$ GHz):** Frequencies must remain well below the superconducting pair-breaking gap of thin-film aluminum ($2\Delta_{\text{Al}} \approx 82$ GHz).

We choose $f_q \approx 5.0$ GHz and $f_r \approx 7.04$ GHz, establishing a large detuning:

$$|\Delta| = |f_q - f_r| \approx 2.04\text{ GHz} \gg g \approx 60\text{ MHz} \tag{8}$$

Operating in this **dispersive regime** prevents the qubit from leaking real photons into the resonator (avoiding Purcell decay). Instead, the qubit dispersively pulls the resonator frequency by $\pm\chi \approx 1$–3 MHz. Probing the phase shift of a transmitted tone at f_r reveals the qubit state without destroying it (Quantum Non-Demolition readout).

3.3 Component Value Calculations

3.3.1 Readout Resonator ($C_r = 340$ fF, $L_r = 1.5$ nH)

A quarter-wave CPW shorted to ground at one end maps to an equivalent parallel LC tank:

$$C_r = \frac{\pi}{4\omega_r Z_0} \approx \frac{\pi}{4 \times (2\pi \times 7.04 \times 10^9) \times 50} \approx 340\text{ fF} \tag{9}$$

$$L_r = \frac{1}{\omega_r^2 C_r} \approx \frac{1}{(2\pi \times 7.04 \times 10^9)^2 \times (340\text{ fF})} \approx 1.50\text{ nH} \tag{10}$$

Checking with the Thompson resonance formula confirms $f_r = 7.042$ GHz.

3.3.2 Transmon Qubit ($C_q = 70$ fF, $L_J = 10$ nH)

Targeting an anharmonicity of $|\alpha| \approx 280$ MHz sets the total shunt capacitance:

$$C_q = \frac{e^2}{2h|\alpha|} \approx 70\text{ fF} \tag{11}$$

To place the transition frequency near 5.0 GHz:

$$\omega_{01} \approx \frac{1}{\sqrt{L_J C_q}} - \frac{E_C}{\hbar} \implies L_J \approx 10.0\text{ nH} \tag{12}$$

This corresponds to a junction critical current of $I_c = \Phi_0/(2\pi L_J) \approx 32.9$ nA.

3.3.3 Coupling Capacitors ($C_{in} = 2$ fF, $C_c = 4$ fF)

- **Input Coupler ($C_{\text{in}} = 2.0$ fF):** Taps power from the feedline into the resonator, setting loaded quality factor $Q_L \approx 7,000$ and photon escape rate $\kappa \approx 1$ MHz.
- **Qubit Coupler ($C_c = 4.0$ fF):** Bridges the resonator and transmon pads, fixing transverse coupling to $g/2\pi \approx 60$ MHz.

3.4 Why Are Grounding and Port Assignments Mandatory?

In high-frequency planar layouts, continuous superconductor film acts as the RF return ground. Without grounding:

1) Circuits possess floating nodes with undefined potentials. In circuit matrix solvers, floating nodes produce singular admittance matrices ($\det(\mathbf{Y}) = 0$).
2) High-frequency return currents find parasitic paths, causing crosstalk.

Port 1 is assigned to the resonator input on the left; Port 2 is assigned to the qubit on the right. This setup lets us measure forward transmission (S_{21}), reflection at the resonator (S_{11}), and reflection at the qubit (S_{22}).

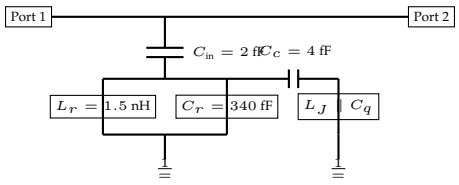


Fig. 6. Lumped-element microwave equivalent network synthesized in Ansys Circuit.

4 Tier-1 Simulation: Ansys Circuit Guided Execution

Full-wave 3D finite-element simulations in Ansys HFSS take 15 to 45 minutes per pass. Modeling the lumped equivalent in Ansys Circuit provides instantaneous feedback (0.1 s), verifying target resonance and bandwidth before opening CAD tools.

4.1 Step-by-Step Circuit Construction

1) **Resonator Tank:** Place $C_5 = 340$ fF in parallel with $L_9 = 1.5$ nH. Ground the bottom rail.
2) **Transmon Tank:** Place $C_3 = 70$ fF in parallel with $L_8 = 10.0$ nH. Ground the bottom rail.
3) **Coupling Network:** Wire $C_4 = 4.0$ fF between the top nodes of both tanks. Wire $C_2 = 2.0$ fF to the top of the resonator tank.
4) **Port Assignments:** Connect Port 1 to the input of C_2. Connect Port 2 directly to the top node of the transmon tank.

4.2 Linear Network Analysis Setup

1) Right-click 'Analysis' in the project tree $\longrightarrow$ 'Add Linear Network Analysis'.
2) Set the frequency sweep from 4.0 GHz to 8.0 GHz with a linear step of 1.0 MHz.
3) Execute 'Analyze All'.
4) Generate Rectangular Reports for S_{11} (resonator dip at 7.04 GHz) and S_{22} (qubit dip near 5.0 GHz).

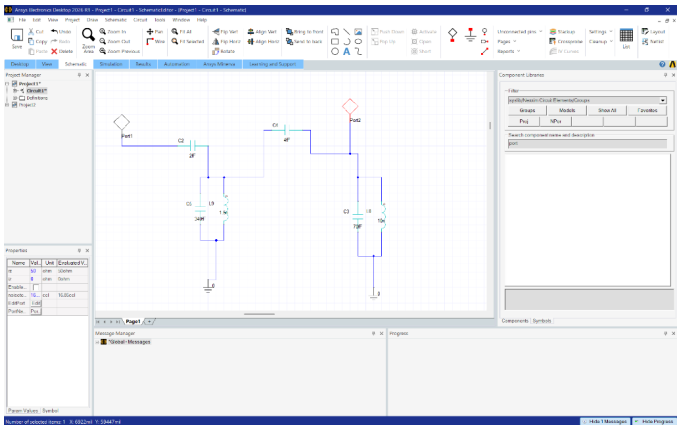


Fig. 7. Synthesized lumped microwave network schematic in Ansys Circuit 2026 R1 with fully wired $50\,\Omega$ ports, coupling capacitors, and ground returns.

APPENDIX A
INTEGRATED THEORETICAL & MICROWAVE LEXICON

A.1 Superconductivity Foundations

Cooper Pairs: Electron pairs coupled below critical temperature (T_c) through electron-phonon interactions, condensing into a single macroscopic ground state with zero electrical resistance.

Superconducting Gap (2Δ): Energy needed to dissociate a Cooper pair into lossy quasiparticles ($2\Delta_{\mathrm{Al}} \approx 82$ GHz).

Kinetic Inductance (L_k): Inductance caused by the inertial mass of accelerating Cooper pairs in thin superconductor traces.

A.2 Josephson Physics

Josephson Junction (JJ): A thin amorphous Al_2O_3 barrier ($\sim$ 1 nm) between two superconductors acting as a nonlinear dissipationless inductor.

Critical Current (I_c): Maximum supercurrent sustainable by a junction before Cooper pairs break.

Josephson Energy (E_J): Characteristic tunneling energy scale: $E_J = \Phi_0 I_c/(2\pi)$.

Josephson Inductance (L_J): Differential inductance: $L_J(\phi) = \Phi_0/(2\pi I_c \cos\phi)$.

A.3 Qubit Regimes

Charging Energy (E_C): Coulomb energy needed to add a Cooper pair: $E_C = e^2/(2C_\Sigma)$.

Cooper-Pair Box: Early charge qubit ($E_J \approx E_C$) plagued by charge-noise dephasing ($T_2^* < 1\,\mu$s).

Transmon Qubit: Circuit operating at $E_J/E_C \gg 1$ (≈ 50), flattening charge dispersion while retaining negative anharmonicity.

Anharmonicity (α): Level spacing non-degeneracy $\alpha = f_{12} - f_{01} \approx -E_C/h$, preventing pulse leakage into higher states.

A.4 Transmission Lines and Resonators

Coplanar Waveguide (CPW): Planar transmission line with a center conductor flanked by ground planes on silicon.

Characteristic Impedance (Z_0): Standardized to $50.0\,\Omega$ to eliminate reflections.

Readout Resonator: Quarter-wave CPW line shorted at one end, storing microwave photons for dispersive state readout.

Quality Factor (Q): Metric of energy storage: $1/Q_L = 1/Q_i + 1/Q_c$.

A.5 Circuit QED Readout

Coupling Rate (g): Vacuum Rabi dipole interaction strength between qubit and cavity.

Dispersive Regime ($|\Delta| \gg g$): Operating space where detuning exceeds coupling, forbidding real photon exchange.

Dispersive Shift (χ): State-dependent resonator frequency shift enabling non-demolition readout.

Purcell Effect: Spontaneous photon emission into feedline modes, mitigated using detuning or bandpass Purcell filters.

A.6 Microwave Theory

Forward Transmission (S_{21}): Transmitted voltage ratio showing the absorption notch at resonance.

Return Loss (S_{11}): Reflected voltage ratio indicating impedance matching.

RF Ground Return: Low-inductance path ensuring high-frequency return currents complete their loops cleanly.

A.7 Cryogenics and Fabrication

Dilution Refrigerator: Cryocooler using ^{3}He/^{4}He dilution to hold 15 mK base temperatures.

Cryogenic Attenuators: Thermalized resistive pads absorbing room-temperature blackbody noise on drive lines.

Two-Level Systems (TLS): Amorphous dielectric defect dipoles that absorb microwave energy and limit T_1.

REFERENCES

[1] J. Koch *et al.*, "Charge-insensitive qubit design derived from the Cooper pair box," *Phys. Rev. A*, vol. 76, p. 042319, 2007.
[2] Z. K. Minev *et al.*, "Energy-participation quantization of artificial quantum systems," *Nature*, vol. 590, pp. 248–254, 2021.
[3] A. Blais *et al.*, "Circuit quantum electrodynamics," *Rev. Mod. Phys.*, vol. 93, p. 025005, 2021.